IoT

Contents

Evaluations

- Theory (CCEE) - 40 Marks
- Lab exam - 40 Marks
- Internal exam - 20 Marks

IoT

- collective network of connected devices and the technology that facilitates communication between devices and the cloud, as well as between the devices themselves
- The Internet of Things integrates everyday “things” with the internet
- The cost of integrating computing power into small objects has now dropped considerably
- these smart objects can automatically transmit data to and from the Internet.
- All these “invisible computing devices” and the technology associated with them are collectively referred to as the Internet of Things.
- A typical IoT system works through the real-time collection and exchange of data.
- An IoT system has three components:
    - Smart devices
        - a device that has been given computing capabilities
        - It collects data from its environment, user inputs, or usage patterns and communicates data over the internet to and from its IoT application.
    - IoT application

        - a collection of services and software that integrates data received from various IoT devices
        - It uses machine learning or artificial intelligence (AI) technology to analyze this data and make informed decisions
        - These decisions are communicated back to the IoT device and the IoT device then responds intelligently to inputs
    - A graphical user interface
        - IoT device can be managed through a graphical user interface
        - eg. mobile applications, web applications

Features

- Artificial Intelligence
    - IoT essentially makes virtually anything “smart”
    - it enhances every aspect of life with the power of data collection artificial intelligence algorithms
- networks
    - Connectivity
    - Networks can exist on a much smaller and cheaper scale while still being practical
    - IoT creates these small networks between its system devices
- Sensors
    - IoT without sensor can not do anything
    - They act as defining instruments which transform IoT from a standard passive network of devices into an active system capable of real-world integration
- Devices
    - IoT exploits purpose-built small devices to deliver its precision, scalability, and versatility

Advantages

- Improved Customer Engagement
    - Current analytics suffer from blind-spots and significant flaws in accuracy; and as noted, engagement remains passive
    - IoT completely transforms this to achieve richer and more effective engagement with audiences
- Technology Optimization
    - The same technologies and data which improve the customer experience also improve device use, and aid in more potent improvements to technology
- Reduced Waste
    - IoT makes areas of improvement clear
    - IoT provides real-world information leading to more effective management of resources
- Enhanced Data Collection
    - IoT makes data collection easier
    - With the collected data we can accurately analyse the world we want
    - It allows an accurate picture of everything

Disadvantages

- Security
    - IoT creates an ecosystem of constantly connected devices communicating over networks
    - The system offers little control despite any security measures
    - This leaves users exposed to various kinds of attackers
- Privacy
    - The sophistication of IoT provides substantial personal data in extreme detail without the user's active participation
- Complexity
    - Some find IoT systems complicated in terms of design, deployment, and maintenance
- Flexibility
    - Many are concerned about the flexibility of an IoT system to integrate easily with another
    - They worry about finding themselves with several conflicting or locked systems
- Compliance
    - Its complexity makes the issue of compliance seem incredibly challenging

pros and cons of IoT

Some of the advantages of IoT include the following:

- Enables access to information from anywhere at any time on any device.
- Improves communication between connected electronic devices.
- Enables the transfer of data packets over a connected network, which can save time and money.
- Collects large amounts of data from multiple devices, aiding both users and manufacturers.
- Analyzes data at the edge, reducing the amount of data that needs to be sent to the cloud.
- Automates tasks to improve the quality of a business's services and reduces the need for human intervention.
- Enables healthcare patients to be cared for continually and more effectively.

Some disadvantages of IoT include the following:

- Increases the attack surface as the number of connected devices grows. As more information is shared between devices, the potential for a hacker to steal confidential information increases.
- Makes device management challenging as the number of IoT devices increases. Organizations might eventually have to deal with a massive number of IoT devices, and collecting and managing the data from all those devices could be challenging.
- Has the potential to corrupt other connected devices if there's a bug in the system.
- Increases compatibility issues between devices, as there's no international standard of compatibility for IoT. This makes it difficult for devices from different manufacturers to communicate with each other.

File

- collection of data/info on secondary storage device
- File = Contents(Data) + Information(Metadata)
- Metadata - File Control Block (inode)
    - type, size, user, group, permissions, time stamp
    - location of data on disk
- Files are stored in File system

- File system is on disk partition, logically divided into 4 sections
    - Boot block, Super block, inode list, data blocks

DBMS - Database Management System

- Is a program that store data, retrieve data and manipulate data
- CRUD operations
    - Create - Insert new records
    - Retrieve - Select existing records
    - Update - Modify existing records
    - Delete - Delete existing records
- Generic program for any data e.g. students, employees exams, financial,...
- e.g. Excel/Spreadsheets

RDBMS - Relational DBMS

- is a generic program that store data, retrieve data and manipulate data ie perform CRUD operations efficiently
- RDBMS organize data into Tables, Rows and Columns. These tables are related to each others --> Relational DBMS
- Built-in relational feature
- Built-in CRUD and other operations
- Less network use (only records)
- Server side processing
- Faster
- huge data support (100s of GB)
- Client-server architecture
- Multi-user systems
- Row level locking
- e.g. MySQL, MariaDb, Oracle, MSSQL, Informix, Sybase, Paradox, SQLite

SQL - Structure Query Language

- RDBMS data processing is done using SQL queries
- Client fire query on server, server process query, produce result and send result back to client
- Case insensitive language
- Categories:
    - DDL : Data Definition Language
        - create, drop, rename, alter
    - DML : Data Manipulation Language *
        - insert, update, delete
    - DQL : Data Query Language *
        - select
    - DCL : Data Control Language
        - create user, grant, revoke
    - TCL : Transaction Control Language
        - commit, rollback

MySQL - RDBMS software

MySQL Installation - Ubuntu

- open the terminal and give following commands.
  - sudo apt-get update
  - sudo apt-get install mysql-server -y
- After Successfull Installation, give following command to open mysql prompt
  - sudo mysql
- Set password of root user as 'root'
  - use mysql;
  - alter user 'root'@'localhost' identified with mysql_native_password by 'root';
  - exit;
- Open Mysql prompt with password
  - mysql -u root -p
    - or
  - mysql -u root -proot
- Create a new user
  - use mysql;
  - CREATE USER 'sunbeam'@'localhost' IDENTIFIED BY 'sunbeam';
- Create database for your usage
  - create database iotdb;
- Grant all permissions for database(iotdb) to user(sunbeam)
  - grant all on iotdb.* to sunbeam@'%';
  - FLUSH PRIVILEGES;
  - exit
- Login with new user and start creating and using tables
- MySQL server - mysqld
  - without UI
  - implemented in C/C++
  - To check status of MySQL server
    - sudo systemctl status mysql
  - To start the service of MySQL server
    - sudo systemctl start mysql
  - To stop the service of MySQL server
    - sudo systemctl stop mysqld

- MySQL client - mysql
  - CLI based
  - To start the MySQL client
    - mysql
    - mysql -u root
    - mysql -u root -p

SQL Queries

- To display all the databases
  - show databases;
- To create a new database
  - create database iotdb;
- To use/change the database
  - use iotdb;
- Person - uid, name, age, address, mobile
- To create a new table
  - create table persons(uid INT, name VARCHAR(36), age INT, address VARCHAR(64), mobile VARCHAR(16));
  - create table persons(uid INT NOT NULL, name VARCHAR(36), age INT, address VARCHAR(64), mobile VARCHAR(16));
  - create table persons(uid INT AUTO_INCREMENT, name VARCHAR(36), age INT, address VARCHAR(64), mobile VARCHAR(16));
- To display all tables
  - show tables;
- To describe the table
  - desc persons;
- To add record into table
  - insert into persons values(42, "abc", 30, "pune", "1234567890");
  - insert into persons(name, uid, age, address, mobile) values("pqr", 65, 25, "mumbai", "9765977654");
  - insert into persons(name, uid, age, address) values("xyz", 60, 27, "mumbai");
  - insert into persons values(41, "abc", 32, "pune", "1234567890"),(34, "mno", 34, "delhi", "9890662093");

- To read records form table
  - select * from persons;
  - select uid from persons;
  - select name from persons;
  - select uid,name from persons;
  - select DISTINCT address from persons;
  - select * from persons where address = "pune";
  - select * from persons where age <= 30;
  - select * from persons where age < 30 or age > 30;
  - select * from persons order by address;
  - select * from persons order by address DESC;
  - select * from persons order by address ASC;
  - select * from persons order by age DESC LIMIT 1;
  - select * from persons order by uid, address;
- To update existing records
  - update persons SET mobile="7658612345" where uid = 60;
  - update persons SET mobile="7658612345", address = "chennai" where uid = 60;
  - update persons SET mobile="7658612345";
- To delete existing records
  - delete from persons where uid = 60;
  - delete from persons;
    - -- all records will be deleted
- To delete all records
  - truncate table persons;
- To delete table
  - drop table persons;
    - -- give error if table does not exist
  - drop table IF EXISTS persons;
- Create table to log data generated by different sensors

    - type
    - location
    - value
    - time

- create table sensorsLog(type VARCHAR(16), location VARCHAR(20), value float, time datetime default CURRENT_TIMESTAMP);
- insert into sensorsLog(type, location, value) values("LM35", "Nira", 39);

Install MySQL connector for python

- sudo pip3 install mysql-connector-python